Hasnat Ahmad

Prospective Master's Researcher | Artificial Intelligence & Data Science

Gujranwala, Punjab, Pakistan | hasnat.ahmad.9593@gmail.com | +92 307 9560172 |
linkedin.com/in/hasnat-ahmad-b99a152b2 | github.com/HAS0786 | hasnat-ahmad-portfolio.vercel.app/

Academic Profile

Information Technology graduate with hands-on expertise in Artificial Intelligence, Python-based Data Science, and robust Software Engineering. Experienced in implementing Retrieval-Augmented Generation (RAG) architectures and integrating Large Language Models (LLMs) into scalable applications. Seeking a Master's research position to contribute to advanced data mining, machine learning, and cognitive computing systems.

Education

Bachelor of Science in Information Technology (BSIT) | 2022 –2026

Government Graduate College Satellite Town, Gujranwala (Affiliated with **Punjab University**)

- **CGPA:** 3.57 / 4.00 | Final result awaited
- Medium of Instruction: English (MOI letter)

Core Technical Skills

- **AI & Machine Learning:** RAG, LLM API integration, prompt engineering, FAISS, sentence embeddings, OCR (Tesseract), Scikit-learn, predictive modeling
- **Data Science & Analytics:** Python (Pandas, NumPy, Matplotlib, Seaborn), EDA, SQL, descriptive statistics, Power BI
- **Software Engineering:** Flutter/Dart, Riverpod, Provider, Flask, Node.js, REST APIs
- **Infrastructure & Cloud:** Firebase Authentication, Cloud Firestore, Supabase, Git/GitHub, Docker, AWS EC2

Projects

Notes House: Intelligent Learning Management System with RAG | Final Year Project

GitHub: github.com/HAS0786/fyp_notes_house | github.com/HAS0786/notes-house-rag-service

- Built a role-based (student, teacher, admin) Flutter learning app with Firebase Authentication, a built-in PDF reader, quizzes and performance tracking; teacher uploads reach students only after admin approval.
- Implemented a RAG chatbot in Flask. multi-format text extraction with OCR fallback, sentence-aware chunking, MiniLM embeddings in FAISS, scoped top-k retrieval and GPT-4o-mini answer generation.
- Added AI quiz generation (editable by teachers) and LLM keyword extraction for note search.

Student Stress Level Prediction: SVM vs Decision Tree | Python, Scikit-learn, Pandas, Seaborn

GitHub: github.com/HAS0786/DataScience_SVM_DecisionTree_Scikit_Project

- Built a 3-class classifier (Low/Medium/High) on about 520 student survey responses, comparing an RBF-kernel SVM (standardized features, class-weighted) with an unpruned Decision Tree.
- Both models performed comparably on a 20% held-out set (accuracy 94% vs 93%, macro-F1 about 0.94); the SVM missed more high-stress cases (recall 0.85 vs 0.92), and its learning curve converged near 0.95 macro-F1, indicating good generalization.

Nova Shop E-Commerce Analysis | Python, Pandas, NumPy

GitHub: github.com/HAS0786/DataScience_Pandas_Project

- Cleaned and analyzed the UCI Online Retail dataset (revenue by country, product, month and hour) and segmented customers with RFM scoring.

Power BI Dashboard Project

GitHub: github.com/HAS0786/DataScience_Pandas_Project

- Designed 3+ interactive Power BI dashboards using DAX measures, Power Query transformations, slicers, and drill-through filters to visualize key business metrics.

Software Engineering Experience

Software Development Intern | CoreLine Logics | Aug 2026 – Present

- Developing Periodly, a cross-platform period-tracking app in Flutter that tracks cycles and calculates upcoming dates with custom date-based logic.
- Managing app state with Riverpod, with Supabase for the database and user authentication.

Software Development Intern | DevelopersHub Corporation | Dec 2025 – Jan 2026

- Built task-based application modules in Dart, working in agile cycles with code reviews.

Certifications

- IBM Data Science Professional Certificate
- IBM Data Analysis with Python
- Microsoft Power BI Professional Certificate

Languages

- Urdu (native)
- English (working proficiency)